Electric Vehicle Market Performance & Insights

Requirements Analysis

Repository: https://github.com/manasgite54-cmd/EV-project

Purpose and Context

The system requirements are separated into functional behavior, quality attributes, data prerequisites and acceptance tests. The checks below are requirements and test intentions, not assertions that every test has already passed.

Detailed Project-Specific Content

- FR-01: load the source dataset and report missing files or unexpected schema clearly.
- FR-02: run documented cleaning/type-conversion rules and export data/processed/ev_market_clean.csv.
- FR-03: create data/processed/cleaning_report.json and reconcile its output count with the CSV.
- FR-04: support manufacturer, model-year and category comparisons where fields exist.
- FR-05: analyze battery type/capacity, range, price and rating only when definitions and units are verified.
- FR-06: provide SQLite storage and SQL queries consistent with the processed dataset.
- FR-07: provide access to the supplied Tableau Dashboard and Story.
- FR-08: expose Overview, Dashboard, Story and Methodology pages through Flask or equivalent implemented routes.
- NFRs: reproducible setup; stable relative paths; clear units; readable/responsive interface; graceful external-embed fallback; maintainable separation of processing, database, routes and templates; and no exposed credentials.
- Data prerequisites: source URL, license, access date, geographic/time coverage, row grain, schema, units and missing-value conventions.
- Acceptance checks: CSV loads; row count reconciles; field mapping is documented; SQL/Pandas results agree for selected checks; routes and embeds work; and no chart label overstates what a field measures.

Project References and Artifact Locations

Project: Electric Vehicle Market Performance & Insights Repository: https://github.com/manasgite54-cmd/EV-project Processed dataset: data/processed/ev_market_clean.csv Reported processed records: 299,999 Cleaning report: data/processed/cleaning_report.json Tableau Dashboard: https://public.tableau.com/views/Book1_17931030431560/EVMarketOverview?:showVizHome=no&:embed=yes Tableau Story: https://public.tableau.com/views/Book1_17931030431560/Story1?:showVizHome=no&:embed=yes Accuracy note: the record count and artifact paths above are reported by the project owner. Exact schema, source provenance, units, calculated statistics, test results, and deployment status must be checked against the repository and actual artifacts. No unsupported results, approvals, or signatures are claimed.

Status: Detailed project-specific document. Any [TO BE PROVIDED], [TO BE VERIFIED] or [TO BE CALCULATED] item requires confirmation. No supervisor approval, signature, unverified test pass or successful deployment is implied.